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Light shielding device for microscope and microscope comprising the same

Abstract

The disclosure relates to a light shielding device for a microscope and a microscope including the same. In the microscope, transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage and fluorescent illumination light from a fluorescent illumination light source is irradiated from below the stage to the specimen on the stage. The light shielding device may be mounted to a lower end portion of a condenser of the microscope to at least partially block ambient light entering an objective of the microscope. The light shielding device may include a light shield and a mounting mechanism configured to mount the light shield below the lower end portion of the condenser in a manually removable manner, and/or the light shield includes a through hole and a light shielding portion surrounding the through hole.

Inventors: **Li; Jing (Shanghai, CN), Dong; Fang (Shanghai, CN), Cao; Junjie (Shanghai, CN), Gu; Xingbiao (Shanghai, CN), Xu; Shukuan (Shanghai, CN)**

Applicant: **CARL ZEISS MICROSCOPY GMBH** (Jena, DE)

Family ID: **1000008818361**

Assignee: **Carl Zeiss Microscopy GMBH (Jena, DE)**

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Primary Examiner: Nguyen; Thong Q

Attorney, Agent or Firm: Armstrong Teasdale LLP

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This patent application is a continuation of U.S. patent application Ser. No. 17/221,504 filed on Apr. 2, 2021, now abandoned, which claims the benefit and priority of Chinese Patent Application No. 202010260711.8 filed on Apr. 3, 2020, the disclosures of which are incorporated by reference herein in their entirety as part of the present application.

TECHNICAL FIELD
(1) The invention relates to the field of microscope, in particular to a microscope with a fluorescence observation function, and more particularly to a light shielding device used for such a microscope.
BACKGROUND ART
(2) Fluorescence microscopes or microscopes with a fluorescence observation function are known in the art. In this type of microscope, a specimen located on a stage of the microscope is irradiated with excitation light from a specified light source when fluorescence observation is performed. The fluorescent substance contained in the specimen is excited by the excitation light to emit fluorescence having a longer wavelength than that of the excitation light, so a fluorescent image of the specimen can be observed through an objective and others of the microscope.
(3) However, the fluorescence is weak light, so if ambient light other than the fluorescence emitted by the specimen also enters the objective of the microscope (in particular, it is easier to receive ambient light when the objective is arranged below the stage), it will mix with the fluorescence and interfere with the fluorescent image of the specimen, making it difficult to obtain a clear fluorescent image with sufficient contrast.
(4) In this regard, it has been proposed in the prior art to equip the microscope with a light shielding device to minimize the ambient light that enters the objective during fluorescence observation of the microscope, so that the contrast of the fluorescent image of the specimen can be improved without needing to operate the microscope in a dark environment. However, the existing light shielding devices still have room for improvement in terms of convenience in use and installation.
(5) The invention aims to overcome one or more defects and/or other problems existing in the prior art.
SUMMARY OF THE INVENTION
(6) In a first aspect, the invention relates to a light shielding device for a microscope, the

microscope comprising a condenser configured so that transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage via the condenser; wherein fluorescent illumination light from a fluorescent illumination light source is irradiated from below the stage to the specimen on the stage; wherein the light shielding device is mounted to a lower end portion of the condenser to at least partially block ambient light entering an objective of the microscope, and wherein the light shielding device comprises a light shield and a mounting mechanism configured to mount the light shield below the lower end portion of the condenser in a manually removable manner.

(7) In a second aspect, the invention relates to a microscope, comprising: a condenser configured so that transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage via the condenser, wherein fluorescent illumination light from a fluorescent illumination light source is irradiated from below the stage to the specimen on the stage; and a light shielding device according to the first aspect.

(8) In a third aspect, the invention relates to a light shielding device for a microscope in which transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage and fluorescent illumination light from a fluorescent illumination light source is irradiated from below the stage to the specimen on the stage; wherein the light shielding device comprises a light shield disposed above the specimen on the stage to at least partially block ambient light entering an objective of the microscope, and wherein the light shield comprises a through hole and a light shielding portion surrounding the through hole, the through hole being configured to allow all or part of the transmitting illumination light to pass through the light shield, and a light shield filter is provided at the through hole.

(9) In a fourth aspect, the invention relates to a microscope, in which transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage and fluorescent illumination light from a fluorescent illumination light source is irradiated from below the stage to the specimen on the stage, and which comprises a light shielding device according to the third aspect.

(10) In a fifth aspect, the invention relates to a microscope, in which transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage and fluorescent illumination light from a fluorescent illumination light source is irradiated from below the stage to the specimen on the stage, and which comprises: a light shielding device disposed above and the specimen on the stage to at least partially block ambient light entering an objective of the microscope; and an illumination light path filter disposed in a light path between the transmitting illumination light source and the specimen on the stage.

(11) The light shielding device for a microscope according to the invention may be mounted to e.g. a lower end portion of the condenser of the microscope to at least partially block ambient light entering the objective, and the light shielding device comprises a light shield and a mounting mechanism configured to mount the light shield below the lower end portion of the condenser in a manually removable manner. In this way, when the microscope is used for fluorescence observation, the light shield featuring a simple structure and easy installation can effectively block the ambient light that would otherwise enter the objective of the microscope, thereby ensuring the contrast and definition of the fluorescent image of the microscope. Moreover, when the microscope is not used for fluorescence observation, the light shield can be quickly and easily removed so as not to hinder other operations (e.g. bright-field observation performed using a transmitting illumination light source) of the microscope. This simplifies the structure of the light shielding device and improves its convenience of use. In addition, according to the invention, a filter, in particular a neutral density filter, may be provided between the transmitting illumination light source and the specimen on the stage of the microscope and/or provided at the light shield. This can not only effectively prevent various ambient light from interfering with the fluorescent image, but

also achieve other observation operations and functions of the microscope even without removing the light shield, which greatly improves the practicality and functionality of the microscope.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other advantages, features, and details of the invention can be derived from the exemplary embodiments described below with reference to the drawings, in which:

(2) FIG. **1** is a perspective view of an exemplary configuration of a microscope according to the invention.

(3) FIG. **2** shows a first embodiment of the light shielding device for a microscope according to the invention.

(4) FIG. **3** shows a second embodiment of the light shielding device for a microscope according to the invention.

(5) FIG. **4** shows a third embodiment of the light shielding device for a microscope according to the invention.

(6) FIG. **5** shows a fourth embodiment of the light shielding device for a microscope according to the invention.

(7) FIG. **6** shows a fifth embodiment of the light shielding device for a microscope according to the invention.

(8) FIG. **7** is a perspective view of another exemplary configuration of a microscope according to the invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(9) FIG. **1** shows an exemplary microscope **100** according to the invention, including some common components such as a base **5**, an arm **8**, a condenser **2**, a stage **3**, an objective **4**, an eyepiece unit **6** and a set of intermediate imaging lenses (not shown). The microscope **100** may be, for example, a so-called inverted microscope, wherein a plurality of objectives **4** having different magnifications and mounted on an objective turret are arranged below the stage **3**, and the objective turret may be rotated to align a selected objective **4** with an aperture **7** in the stage **3** as needed, in order to observe a specimen (not shown) placed above the aperture **7** on the stage **3**. The condenser **2** is, for example, cylindrical as a whole and is mounted vertically at a tip end of the arm **8**. The condenser **2** is configured such that transmitting illumination light (for example, white light) from a transmitting illumination light source **1** (for example, including a light emitting diode, which is generally a white light LED) is collected by the condenser **2** (more specifically, converged by one or more condenser lenses included in the condenser **2**), then is irradiated from above the stage **3** to the specimen on the stage, and thereafter enters the objective **4** through the specimen, thereby producing a bright field image to enable bright field observation. In the illustrated embodiment, the transmitting illumination light source **1** is also mounted to the arm **8**, but the transmitting illumination light source may be an external light source provided outside the microscope. The microscope **100** also has a fluorescence observation function. To this end, the microscope **100** further includes a fluorescent illumination light source (not shown, generally multi-channel monochromatic LEDs) that can be built inside or outside the base **5** of the microscope. Certain monochromatic illumination light emitted from the fluorescent illumination light source becomes excitation light (for example, ultraviolet light) after passing through an excitation filter, and then is irradiated from below the stage **3** through the objective **4** to the specimen on the stage, thereby exciting the fluorescent substance contained in the specimen to emit fluorescence. After the fluorescence is reflected by the specimen, it travels downward to enter the objective **4**, and finally into the eyepiece unit **6** through the set of intermediate imaging lenses. In this way, fluorescence observation is realized and a fluorescent image of the specimen can be observed.

(10) In the fluorescence observation mode of the microscope **100**, in order to prevent ambient light from entering the objective **4** and mixing with the fluorescence emitted from the specimen to reduce the contrast of the fluorescent image, a light shielding device **20** is mounted between the transmitting illumination light source **1** and the specimen on the stage, e.g. to an lower end portion of the condenser **2** to at least partially block the ambient light that would otherwise enter the objective **4**. Thus fluorescence observation can be performed with the microscope **100** even if it is not in a dark environment.
(11) According to an embodiment of the invention, the light shielding device **20** comprises a light shield **21** and a mounting mechanism configured to mount the light shield below the lower end portion of the condenser **2** in a manually removable manner. Herein, “mount . . . in a manually removable manner” means that the light shield can be easily and quickly attached to and detached from the condenser **2** by hand without using a tool. FIGS. **2** to **5** show several exemplary configurations of the mounting mechanism.
(12) In the embodiment shown in FIG. **2**, the mounting mechanism includes a collar **22** mounted around the lower end portion of the condenser **2**. To this end, the collar **22** may be in the form of, for example, a circular ring corresponding to the circular cross section of the condenser **2**. A plurality of threaded holes **26** (e.g. three are illustrated in the figure), which are preferably evenly spaced apart from each other, may be provided circumferentially in the collar **22**, and a lock screw **25** is fitted in each of the threaded holes **26**. By screwing the lock screws **25** radially inward into the corresponding threaded holes **26**, the tip end of each lock screw **25** is abutted against the outer circumferential surface of the condenser **2**, thereby fixing the collar **22** to the lower end portion of the condenser **2**. The vertical position of the collar **22** may be adjusted by loosening each lock screw **25**, then moving the collar **22** upward or downward to a desired new position, and then tightening each lock screw. It may be understood by those skilled in the art that the manner of mounting the collar **22** to the lower end portion of the condenser **2** is not limited to use of the above-mentioned lock screws **25**; instead, various ways of connecting, such as adhesive bonding, threaded connection (by providing matching threads on the outer circumferential surface of the condenser **2** and the inner circumferential surface of the collar **22**), and form-fit connection (e.g. snap-fit) may be adopted as appropriate. The collar **22** may also be configured to be integrally formed with the casing of the condenser **2** at the lower end portion of the condenser **2**.
(13) One or more magnets **27** are fixed to the lower end surface of the collar **22** by e.g. adhesive bonding, and the light shield **21** may be made at least partially by a ferromagnetic material. Thus, the light shield **21** can be conveniently attached to the collar **22** by the magnetic attractive force with the magnets **27**, and can be removed easily and quickly from the collar **22** by only overcoming the magnetic attractive force by hand. In order to facilitate the disassembly and assembly of the light shield **21** by hand. A bending portion **28** for hand grasping may be provided at the peripheral edge of the light shield **21**. In the illustrated embodiment, six magnets **27** are evenly distributed in the circumferential direction, but the invention is obviously not limited to this. The number of the magnets **27** may vary depending on the weight of the light shield **21** and the applied force desired to detach the light shield **21**. In addition, in order to quickly and accurately mount the light shield **21** in place to achieve a good light shielding effect, a depression **24** may be provided in the upper surface of the light shield **21**. When the light shield **21** is attached to the collar **22**, the lower end portion of the collar **22** is received in the depression **24** in such a manner that the depression **24** can define the position of the light shield **21** relative to the collar **22** through the engagement between the inner side wall of the depression **24** and the outer circumferential wall of the lower end portion of the collar **22**.
(14) The height of the light shield **21** or the distance between it and the stage **3** may be set by adjusting the vertical position of the collar **22** on the condenser **2** and/or by adjusting the axial length (i.e. width in vertical direction shown in FIG. **2**) of the collar **22**. A desired light shielding effect can be achieved by appropriately setting the height and/or size of the light shield **21**

according to the space arrangement requirements in the microscope **100**. The light shield **21** may be a solid thin plate, but may also be configured as a closed plate including a through hole **23** and a light shielding portion surrounding the through hole **23** as shown in FIG. **2**. The through hole **23** is positioned, sized and shaped to allow all or part of the transmitting illumination light from the condenser **2** to pass through the light shield **21**. During fluorescence observation, since the edge of the through hole **23** in the light shield **21** may be arranged e.g. in close proximity to the lower end portion of the condenser **2**, no ambient light is allowed to pass downward through the through hole **23**, while other ambient light can still be blocked by the light shielding portion surrounding the through hole **23**. When it is switched from the fluorescence observation mode to the bright field observation mode performed using the transmitting illumination light source **1**, the transmitting illumination light can still be radiated downward from the condenser **2** through the through hole **23** without being blocked by the light shield **21**. That is, the microscope can be switched from the fluorescence observation mode using the fluorescent illumination light source to the bright field observation using the transmitting illumination light source **1**, without removing the light shield **21**, which improves the use convenience of the microscope **100**.
(15) In the case where the light shield **21** is provided with the through hole **23**, during fluorescence observation, the excitation light that is irradiated upward to the specimen on the stage through the objective **4** will pass through the normally translucent specimen and continue to travel upward to the LED fluorescent powder in the transmitting illumination light source, which will be excited to generate auto-fluorescence. The auto-fluorescence can be irradiated downward and enter the objective **4**, thereby disturbing the fluorescent image of the specimen. In order to reduce the adverse effect of the auto-fluorescence, the microscope **100** according to the invention may be provided with a filter (hereinafter referred to as “illumination light path filter”) in a light path between the transmitting illumination light source and the specimen on the stage. For example, an illumination light path filter **9** may be arranged in a light path between the transmitting illumination light source **1** and the condenser **2**, as shown in FIG. **1**. More specifically, the illumination light path filter **9** may be disposed e.g. at a light outlet of the transmitting illumination light source **1**. The illumination light path filter **9** may be received, for example, in a slider **10** and inserted between the light source **1** and the condenser **2** by way of the slider **10**.
(16) Such an illumination light path filter can selectively reduce transmission ratio of certain spectrum range, and may be any one of a neutral density filter, a color filter, a bandpass filter, a notch filter, a long pass filter, a spectral filter and a multivariate optical element that can tune transmission at different wavelengths.
(17) Preferably, the illumination light path filter has different transmissions for light of different wavelengths to match with emission spectral intensity of the transmitting illumination light source. Transmission of the illumination light path filter may be inversely proportional to the emission spectral intensity of the transmitting illumination light source. Transmission of the illumination light path filter may be lower for a wavelength with a higher emission spectral intensity, and higher for a wavelength with a lower emission spectral intensity. The maximum transmission may be not higher than 2%, while the minimum transmission may be not lower than 1%.
(18) In view of the fact that visual observation requires light of different wavelengths is tuned at the same ratio after passing through the filter in order to avoid color cast, the illumination light path filter **9** is preferably a neutral density filter. The neutral density filter **9** can attenuate light of different wavelengths passing therethrough in the same proportion/ratio, and thus the excitation light radiated upward through the specimen will be attenuated by the neutral density filter **9**, so the auto-fluorescence excited by the attenuated excitation light in the LED of the transmitting illumination light source will be reduced. The auto-fluorescence will be attenuated again by the neutral density filter **9** when it is irradiated downward. Eventually, little or no auto-fluorescence enters the objective **4**, thereby helping to improve the contrast of the fluorescent image. In addition to this advantage, the neutral density filter **9** also enables normal use of phase contrast, differential

interference contrast (DIC), plastic differential interference contrast (Plas-DIC) and other observation methods of the microscope under transmitting light illumination without user operations (unnecessary to remove the light shield **21**), and helps to realize the function of capturing fluorescent and bright field images with one button click as well as the function of switching between fluorescence and bright field observations with one button click.

(19) As an example, the following optical parameters may be adopted for the illumination light path filter **9** that is preferably a neutral density filter: an average reflection of not more than 4% for light with a wavelength of 420 nm to 680 nm, an average transmission of not more than 2% (corresponding to an optical density of not less than 1.7) for light with a wavelength of 350 nm to 400 nm, and an average transmission between 1.3% and 2% (corresponding to an optical density between 1.7 and 1.9) for light with a wavelength of 400 nm to 700 nm.

(20) It is also conceivable for those skilled in the art that an opaque shield may be used in place of the illumination light path filter **9** to completely block the light path (having the same effect as the case where the light shield **21** is a solid thin plate) in consideration of cost saving etc. However, the opaque shield will block not only the upward excitation light and the downward auto-fluorescence, but also the transmitting illumination light radiated downward. Therefore, it is not possible to realize the function of capturing fluorescent and bright field images with one button click and the function of switching between fluorescence and bright field observations with one button click. If a bright field image produced by using transmitting illumination light is desired, the user needs to push away the slider **10**, and in this case the convenience of use will be impaired.

(21) In the embodiments shown in FIGS. **3** and **4**, the mounting mechanism still includes a collar **22** which can be mounted around the lower end portion of the condenser **2** in the same way as the collar **22** shown in FIG. **2**.

(22) However, a protrusion **33** is provided on the lower end portion of the collar **22** shown in FIG. **3**. The protrusion **33** has, for example, a semicircular shape or a minor arc shape extending in the circumferential direction of the collar **22** and has an L-shaped cross section. In this way, a slot **34** having a semicircular shape or a minor arc shape as a whole is formed between the two branches of the L shape. Correspondingly, a circular boss **31** is provided on the upper surface of the light shield **21**, and a circular or arc-shaped flange **32** is formed along the circumferential direction of the circular boss **31**. The flange **32** is configured to be inserted and fitted into the slot **34** in a direction indicated by the blank arrow shown in FIG. **3**, and thereby the light shield **21** can be conveniently mounted to the lower end portion of the collar **22**. The light shield **21** can be removed easily as long as the flange **32** is simply pulled out of the slot **34** in a reverse direction. Those skilled in the art can easily understand that positions of the slot and the flange may be exchanged, that is, the slot is arranged on the light shield **21** and the flange is arranged on the collar **22**. In addition, the shape of the slot and the flange is not limited to a circular shape or an arc shape, but may be, for example, a linear shape. Naturally, in the case of a linear shape, it is preferable that the slot and the flange are arranged at positions that do not block the hollow portion of the collar **22** or the opening **23** (if provided) of the light shield **21**.

(23) The upper surface of the light shield **21** shown in FIG. **4** is also provided with a circular boss **51**, which is formed with a plurality of radially protruding protrusions **52** arranged at intervals in the circumferential direction (in the embodiment of FIG. **4**, three evenly spaced protrusions are provided, but only two of them are shown). The lower end portion of the collar **22** shown in FIG. **4** is provided with a plurality of L-shaped grooves in one-to-one correspondence to the plurality of protrusions **52**. Specifically, each L-shaped groove is composed of a first portion **53** extending vertically and a second portion **54** extending horizontally. When the light shield **21** is mounted to the collar **22**, each of the protrusions **52** is first aligned with the first portion **53** of a corresponding L-shaped groove, and then the light shield **21** is moved upward so that the protrusion **52** is lifted a sufficient distance in the corresponding first portion **53**. Next, the light shield **21** is rotated in the direction of the black arrow indicated in FIG. **4**, so that the protrusion **52** is rotated in the horizontal

plane into the second portion **54** of the corresponding L-shaped groove, and is thereby supported on a portion of the collar delimiting the second portion **54**. It can be seen that a bayonet connection is thus formed between the collar **22** and the circular boss **51**. Those skilled in the art can readily understand that the bayonet connection can be easily disengaged by performing the above actions in a reverse order. In addition, positions of the protrusions **52** and the L-shaped grooves may be exchanged. For example, protrusions protruding radially inward may be provided on the inner circumferential surface of the lower end portion of the collar **22**, while corresponding L-shaped grooves may be provided on the outer circumferential surface of the circular boss **51**. In order to facilitate disassembly, two recessed portions **29** facing each other may be provided on the outer peripheral edge of the light shield **21** to facilitate grasping by two fingers of a user hand.
(24) In the embodiment shown in FIG. **5**, the mounting mechanism includes at least two clamping members **41** and **42** biased toward each other in e.g. a resilient manner (may be realized by e.g. a tensile spring connected therebetween, not shown) to clamp the lower end portion of the condenser **2**. To this end, the radially inwardly facing surfaces of the clamping members **41**, **42** may be designed as arc-shaped surfaces to match and frictionally engage the outer circumferential surface of the condenser **2**. Grips **43**, **44** may be provided on the clamping members **41** and **42**, respectively, and both hands may grasp them to pull apart the two clamping members against the spring force. In this way, the clamping members **41** and **42** can clamp the condenser **2** and be removed from it. The light shield **21** can be fixed to the lower end surface of one of the at least two clamping members by e.g. adhesive bonding or any other suitable methods, so that it can be mounted and dismounted together with the clamping members.
(25) In the embodiments shown in FIGS. **3** to **5**, the light shield **21** may optionally include the through hole **23** as described above, so that it is unnecessary to remove the light shield **21** from the condenser **2** during bright field observation of the microscope.
(26) It has been described above that the illumination light path filter **9** arranged in the light path between the transmitting illumination light source **1** and the specimen on the stage is used to favorably attenuate the excitation light and auto-fluorescence passing through the through hole **23**. As an alternative or additional component to the illumination light path filter **9**, as shown in FIG. **6**, another filter **60** (hereinafter referred to as “light shield filter”) may be provided at the through hole **23** of the light shield **21** to close the through hole. The effect of the light shield filter **60** is the same as that of the illumination light path filter **9** shown in FIG. **1**. The light shield filter **60** may be attached in the through-hole **23** or attached to the edge of the through-hole **23** by, for example, adhesive bonding. Alternatively, as shown in FIG. **6**, a holder **61** may be provided in the through hole **23**, and a depression **63** may be provided in the holder **61** to receive the light shield filter **60**. The holder **61** may be fixed at the edge of the through-hole **23** through, for example, a plurality of fasteners **62**, thereby the light shield filter **60** is indirectly arranged at the through-hole **23**. The light shield filter **60** may have the same type and optical properties as the illumination light path filter **9** described above (for example, the filters **9** and **60** may be identical neutral density filters).
(27) During transmitting illumination, the transmitting illumination light radiated downward will also be attenuated by the illumination light path filter **9** and/or light shield filter **60** described above. For this, the driving current of e.g. the white light LED in the transmitting illumination light source may be set to maximum the light intensity so as not to impair the bright field effect of the transmitting illumination light source. Besides, optical parameters of the filter(s) may be set by comprehensively taking account of the energy of the existing light sources and acceptable bright field effect, such that the attenuation by the filter(s) would not adversely affect the bright field effect.
(28) A person skilled in the art can appreciate that either or both of the illumination light path filter **9** and the light shield filter **60** may be provided in the microscope system depending on the specific product design and/or application. It would also be appreciated by those skilled in the art that the terms “illumination light path filter **9**” and “light shield filter **60**” as used herein are just for the

purpose of differentiation between the two filters according to their respective positions in the microscope system, and both of them may be any conventional filter in the field of optical components, such as a neutral density filter, a color filter, a bandpass filter, a notch filter, a long pass filter, a spectral filter or a multivariate optical element that can tune transmission at different wavelengths.

(29) According to the configurations of the light shielding devices **20** shown in FIGS. **2** to **5**, a simple light shield **21** can be conveniently mounted to the lower end portion of the condenser **2** in the fluorescence observation mode to effectively block ambient light; and when fluorescence observation is not performed, the light shield **21** can be quickly and easily removed manually, which simplifies the structure of the light shielding device and improves the convenience of use. As shown in FIG. **7**, the light shield **21** removed from the condenser **2** may be placed, for example, in a recess (not shown) provided on the bottom surface of the base **5** of the microscope **100** in order not to be lost. The bending portion **28** provided on the light shield **21** helps to pull the light shield **21** from the recess and mount it onto the condenser **2** again.

(30) The embodiments disclosed above are merely examples of the invention, and all obvious modifications and variations made by those skilled in the art should be considered as falling within the scope of the invention.

Claims

1. A light shielding device for a microscope, the microscope comprising a condenser configured so that transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage via the condenser, wherein fluorescent illumination light is irradiated from below the stage to the specimen on the stage, wherein the light shielding device is mounted to a lower end portion of the condenser to at least partially block ambient light entering an objective of the microscope, wherein the light shielding device comprises a light shield and a mounting mechanism configured to mount the light shield below the lower end portion of the condenser in a manually removable manner, wherein the light shield comprises a through hole and a light shielding portion surrounding the through hole, the through hole is configured to allow all or part of the transmitting illumination light from the condenser to pass through the light shield, wherein a light shield filter is provided at the through hole, the light shield filter is a neutral density filter, and wherein the neutral density filter attenuates light of different wavelengths passing therethrough in the same proportion/ratio.

2. The light shielding device according to claim 1, wherein the mounting mechanism comprises a collar mounted around or integrally formed with the lower end portion of the condenser, wherein one or more magnets are fixed on a lower end surface of the collar, and wherein the light shield contains a ferromagnetic material and is attached to the collar by a magnetic attractive force between the magnet(s) and the ferromagnetic material.

3. The light shielding device according to claim 2, wherein a depression is provided in an upper surface of the light shield wherein the light shield is attached to the collar, a lower end portion of the collar is received in the depression to define a position of the light shield relative to the collar.

4. The light shielding device according to claim 1, wherein the mounting mechanism comprises a collar mounted around or integrally formed with the lower end portion of the condenser, wherein one of a lower end portion of the collar and an upper surface of the light shield is provided with a slot, wherein and the other of the lower end portion of the collar and the upper surface of the light shield is provided with a flange fitted in the slot, or the mounting mechanism comprises a collar mounted around or integrally formed with the lower end portion of the condenser, wherein a circular boss is provided on an upper surface of the light shield, wherein one of a lower end portion of the collar and the circular boss is formed with a plurality of radially protruding protrusions arranged circumferentially at intervals, wherein the other of the lower end portion of the collar and

the circular boss is formed with a plurality of L-shaped grooves corresponding to the plurality of protrusions, and wherein the plurality of protrusions are respectively received in the plurality of L-shaped grooves to form a bayonet connection between the collar and the circular boss, or the mounting mechanism comprises at least two clamping members resiliently biased toward each other to clamp the lower end portion of the condenser, and wherein the light shield is fixed to a lower end surface of one of the at least two clamping members.

5. The light shielding device according to claim 1, wherein the light shield filter has an average reflection of not more than 4% for light with a wavelength of 420 nm to 680 nm, an average transmission of not more than 2% for light with a wavelength of 350 nm to 400 nm, and an average transmission between 1.3% and 2% for light with a wavelength of 400 nm to 700 nm.

6. A microscope comprising: a condenser configured so that transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage via the condenser, wherein fluorescent illumination light is irradiated from below the stage to the specimen on the stage; and a light shielding device according to claim 1.

7. The microscope according to claim 6, wherein a recess is provided on a bottom surface of a base of the microscope to receive the light shield removed from the condenser.

8. The microscope according to claim 6, wherein the microscope further comprises an illumination light path filter disposed in a light path between the transmitting illumination light source and the specimen on the stage.

9. The microscope according to claim 8, wherein the illumination light path filter is disposed at a light outlet of the transmitting illumination light source, or the illumination light path filter is any one of a neutral density filter, a color filter, a bandpass filter, a notch filter, a long pass filter, a spectral filter and a multivariate optical element that can tune transmission at different wavelengths, or the illumination light path filter has different transmissions for light of different wavelengths to match with emission spectral intensity of the transmitting illumination light source, wherein transmission of the illumination light path filter is inversely proportional to the emission spectral intensity of the transmitting illumination light source, and wherein transmission of the illumination light path filter is lower for a wavelength with a higher emission spectral intensity, and is higher for a wavelength with a lower emission spectral intensity, or the illumination light path filter has an average reflection of not more than 4% for light with a wavelength of 420 nm to 680 nm, an average transmission of not more than 2% for light with a wavelength of 350 nm to 400 nm, and an average transmission between 1.3% and 2% for light with a wavelength of 400 nm to 700 nm.

10. A light shielding device for a microscope in which transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage and fluorescent illumination light is irradiated from below the stage to the specimen on the stage, wherein the light shielding device comprises a light shield disposed above the specimen on the stage to at least partially block ambient light entering an objective of the microscope, wherein the light shield comprises a through hole and a light shielding portion surrounding the through hole, the through hole is configured to allow all or part of the transmitting illumination light to pass through the light shield, and a light shield filter is provided at the through hole, wherein the light shield filter is a neutral density filter, and wherein the neutral density filter attenuates light of different wavelengths passing therethrough in the same proportion/ratio.

11. The light shielding device according to claim 10, wherein the light shield filter has an average reflection of not more than 4% for light with a wavelength of 420 nm to 680 nm, an average transmission of not more than 2% for light with a wavelength of 350 nm to 400 nm, and an average transmission between 1.3% and 2% for light with a wavelength of 400 nm to 700 nm.

12. The light shielding device according to claim 10, wherein the microscope comprises a condenser configured so that the transmitting illumination light from the transmitting illumination light source is irradiated from above the stage to the specimen on the stage via the condenser, the

light shielding device is mounted to a lower end portion of the condenser and further comprises a mounting mechanism configured to mount the light shield below the lower end portion of the condenser in a manually removable manner.

13. The light shielding device according to claim 12, wherein the mounting mechanism comprises a collar mounted around or integrally formed with the lower end portion of the condenser, wherein one or more magnets are fixed on a lower end surface of the collar, and wherein the light shield contains a ferromagnetic material and is attached to the collar by a magnetic attractive force between the magnet(s) and the ferromagnetic material.

14. The light shielding device according to claim 13, wherein a depression is provided in an upper surface of the light shield wherein the light shield is attached to the collar, a lower end portion of the collar is received in the depression to define a position of the light shield relative to the collar.

15. A microscope, in which transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage and fluorescent illumination light is irradiated from below the stage to the specimen on the stage, and which comprises a light shielding device according to claim 10.

16. A microscope, in which transmitting illumination light from a transmitting illumination light source is irradiated from above a stage to a specimen on the stage and fluorescent illumination light is irradiated from below the stage to the specimen on the stage, and which comprises: a light shielding device disposed above and the specimen on the stage to at least partially block ambient light entering an objective of the microscope; an illumination light path filter disposed in a light path between the transmitting illumination light source and the specimen on the stage, wherein the illumination light path filter is a neutral density filter, and wherein the neutral density filter attenuates light of different wavelengths passing therethrough in the same proportion/ratio.

17. The microscope according to claim 16, wherein the illumination light path filter is disposed at a light outlet of the transmitting illumination light source, or the illumination light path filter has an average reflection of not more than 4% for light with a wavelength of 420 nm to 680 nm, an average transmission of not more than 2% for light with a wavelength of 350 nm to 400 nm, and an average transmission between 1.3% and 2% for light with a wavelength of 400 nm to 700 nm.

18. The microscope according to claim 16, wherein the microscope further comprises a condenser configured so that the transmitting illumination light from the transmitting illumination light source is irradiated from above the stage to the specimen on the stage via the condenser, the light shielding device is mounted to a lower end portion of the condenser and comprises a light shield and a mounting mechanism configured to mount the light shield below the lower end portion of the condenser in a manually removable manner.

19. The microscope according to claim 18, wherein the mounting mechanism comprises a collar mounted around or integrally formed with the lower end portion of the condenser, wherein one or more magnets are fixed on a lower end surface of the collar, and wherein the light shield contains a ferromagnetic material and is attached to the collar by a magnetic attractive force between the magnet(s) and the ferromagnetic material.

20. The microscope according to claim 19, wherein a depression is provided in an upper surface of the light shield wherein the light shield is attached to the collar, a lower end portion of the collar is received in the depression to define a position of the light shield relative to the collar.

21. The microscope according to claim 18, wherein the mounting mechanism comprises a collar mounted around or integrally formed with the lower end portion of the condenser, wherein one of a lower end portion of the collar and an upper surface of the light shield is provided with a slot, and wherein the other of the lower end portion of the collar and the upper surface of the light shield is provided with a flange fitted in the slot, or the mounting mechanism comprises a collar mounted around or integrally formed with the lower end portion of the condenser, wherein a circular boss is provided on an upper surface of the light shield, wherein one of a lower end portion of the collar and the circular boss is formed with a plurality of radially protruding protrusions arranged

circumferentially at intervals, wherein the other of the lower end portion of the collar and the circular boss is formed with a plurality of L-shaped grooves corresponding to the plurality of protrusions, and wherein the plurality of protrusions are respectively received in the plurality of L-shaped grooves to form a bayonet connection between the collar and the circular boss, or the mounting mechanism comprises at least two clamping members resiliently biased toward each other to clamp the lower end portion of the condenser, and the light shield is fixed to a lower end surface of one of the at least two clamping members, or the light shield comprises a through hole and a light shielding portion surrounding the through hole, the through hole is configured to allow all or part of the transmitting illumination light from the condenser to pass through the light shield, or a recess is provided on a bottom surface of a base of the microscope to receive the light shield removed from the condenser.